

Algebraic geometry 2026

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September 16, 2026

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Chapter 1

Mynotes from 2026 class

Some references used here:

- Mumford's red book [3]
- Humphreys Linear algebraic groups [2]

1.1 Lecture 1 and 2

First some definitions and lemmas worth to remember.

- (Integral extension). Let R be a ring and $S \subset R$ a subring. We say that R is integrally dependent over S if every $r \in R$ is integral over S , that is, there exist a monic polynomial $f \in S[X]$ such that $f(r) = 0$.

Lemma 1.1.1. Let $S \subset R$ be rings. A finite subset of elements $b_1, \dots, b_m \in R$ are all integral over S if and only if $S[b_1, \dots, b_m]$ is finitely generated (as a S -module) over S .

Proof: ($\Rightarrow$) Let $b \in R$ be integral over S , i.e., there exist

$$f = X^d + a_1X^{d-1} + \dots + a_d \in S[X] \quad (1.1)$$

such that $f(b) = 0$. Then, using long division, we see that $S[b] = S + \dots + Sb^{d-1}$ which is finitely generated. Now, if $b_1, \dots, b_m$ are all integral over S , we can argue by induction that $S[b_1, \dots, b_{m-1}]$ is finitely generated over S . However, since b_m is integral over $S[b_1, \dots, b_{m-1}]$, $S[b_1, \dots, b_m]$ is finitely generated over $[b_1, \dots, b_{m-1}]$ and it must be finitely generated over S aswell.

($\Leftarrow$) Suppose that $S[b_1, \dots, b_m] = S\beta_1 + \dots + S\beta_n$ for some $\beta_i \in R$. Then for any $b \in R$ which fixes $S[b_1, \dots, b_m]$, we have $b\beta_i = \sum_j a_{ij}\beta_j$ for some $(a_{ij}) \in \text{Mat}_n(S)$. Therefore

$$0 = (bI - (a_{ij}))^*(bI - (a_{ij}))v = \det(bI - (a_{ij}))v \quad (1.2)$$

where $v = (\beta_1, \dots, \beta_n)^T$. Therefore $\det(bI - (a_{ij}))\beta_i = 0, \forall i$ and since $1 \in S\beta_1 + \dots + S\beta_n$, we see that $\det(bI - (a_{ij})) = 0$. Since $\det(X - (a_{ij})) \in S[X]$ is monic, we are done. $\square$

Corollary 1.1.2. (Transitive property of integral dependence). Given rings $R \supset S \supset T$, if R is integral over S and S is integral over T , then R is integral over T .

Question: Do we need Noetherian hypothesis using this statement?

Proof: Let $r \in R$, then by hypothesis there exist $f = X^d + s_1X^{d-1} + \dots \in S[X]$ such that $f(r) = 0$. Define $A = T[s_1, \dots, s_d]$, then $A[r]$ is a finitely generated A -module because $S \subset A$ and lemma (1.1.1). Now, since A is a finitely generated T -module by lemma (1.1.1), we see that $A[r]$ is a finitely generated T -module. Now, **assuming T is Noetherian** we see that $T[r]$ is finitely generated as a T -module. $\square$

Remark 1.1.3. We can certainly say (using the previous proof) that if $r \in R$, then $S[r]$ is finitely generated as a T -module.

Lemma 1.1.4. Let $K \supset k$ (k a perfect field) be a finitely generated extension field with transcendence degree $\partial_k(K) = d$, then there exist generators $z_1, \dots, z_d, z_{d+1}$ such that

- (1) $K = k(z_1, \dots, z_{d+1})$
- (2) $\{z_1, \dots, z_d\}$ are algebraically independent
- (3) z_{d+1} is separable over $k(z_1, \dots, z_d)$

Lemma 1.1.5. (Noether's normalization lemma - Mumford's red book). Let R be a finitely generated algebra over k . Assume also that R is an integral domain. If R has transcendence degree n over k (which means that $\partial_k \text{Frac}(R) = n$), then there exist $x_1, \dots, x_n \in R$ algebraically independent and such that R is integral over $k[x_1, \dots, x_n]$.

TODO: Study prove this version of the lemma using the statement made in class.

Lemma 1.1.6. (Noether's normalization lemma - Statement made in class). Let R be a finitely generated k -algebra. Then there exist $x_1, \dots, x_n \in R$ algebraically independent over k such that R is integral over $k[x_1, \dots, x_n]$.

Proof (Nagata): Since R is finitely generated as a k -algebra, there exist generators $y_1, \dots, y_m$ such that $R = k[y_1, \dots, y_m]$. Now suppose that $y_1, \dots, y_m$ are algebraically independent, then we can take $x_i = y_i$. If not, then there exist $p \in k[Y_1, \dots, Y_m] \setminus 0$ such that $p(y_1, \dots, y_m) = 0$. Write

$$p(Y_1, \dots, Y_m) = \sum_{\alpha \in \mathbb{Z}_{\geq 0}^m} a_\alpha Y_1^{\alpha_1} \dots Y_m^{\alpha_m} \quad (1.3)$$

and define a new polynomial

$$\begin{aligned} q(Y_1, \dots, Y_m) &= p(Y_1, Y_2 + Y_1^{r_2}, \dots, Y_m + Y_1^{r_m}) \\ &= \sum_{\alpha} a_{\alpha} (Y_1^{\alpha_1 + r_2 \alpha_2 + \dots + r_m \alpha_m} + \dots) \end{aligned} \quad (1.4)$$

for some choice of positive integers $r_2, \dots, r_m$. Take a positive number $b > \alpha_i, \forall \alpha_i$ such that $a_{\alpha} \neq 0$. If we take $r_i = b^i$, then $\alpha_1 + \alpha_2 b + \dots + \alpha_m b^{m-1}$ are all different numbers in base b . So, from different powers of the form $\alpha_1 + r_2 \alpha_2 + \dots + r_m \alpha_m$ we have a maximal one $d = \max\{\alpha_1 + \dots + r_m \alpha_m : a_{\alpha} \neq 0\}$ and

$$q(Y_1, \dots, Y_m) = aY_1^d + \text{smaller degree terms} \in k[Y_2, \dots, Y_m][Y_1], \quad a \neq 0. \quad (1.5)$$

Now, if we let $z_i = y_i - y_1^{r_i} \in R, i = 2, \dots, m$, we see that $q(y_1, z_2, \dots, z_m) = p(y_1, \dots, y_m) = 0$ and y_1 is integral over $k[z_2, \dots, z_m]$. By induction on number of generators, there exist $x_1, \dots, x_n$ algebraically independent such that $k[z_2, \dots, z_m]$ is integral over $k[x_1, \dots, x_n]$. So, by lemma 1.1.1 and its corollary (1.1.1), we see that $k[y_1, z_2, \dots, z_m]$ is integral over $k[x_1, \dots, x_m]$. Now, by definition, it's clear that $k[y_1, z_2, \dots, z_m]$ is a finitely generated R -module. So by the corollary (1.1.2) again, we see that R is integral over $k[x_1, \dots, x_m]$. $\square$

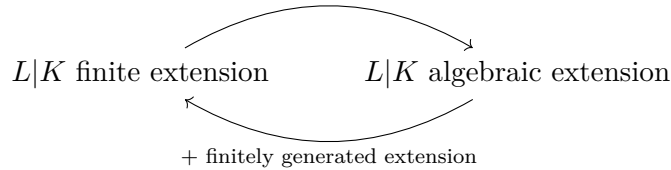
Lemma 1.1.7. Let $S \subset R$ be a ring extension. If R is a field, which integral over S , then S is a field.

Proof: Let $s \in S \setminus 0$ and $r \in R$ be its inverse. Since R is integral over S , there exist $f = X^d + a_1 X^{d-1} + \dots \in S[X]$ be such that $f(r) = 0$. Therefore

$$0 = s^{d-1} f(r) = r + a_1 + sa_2 + \dots + s^{d-1} a_0 \Rightarrow r \in S. \quad (1.6)$$

$\square$

Remark 1.1.8. Let $L|K$ be a field extension. Then



Theorem 1.1.9. (Zariski). Let k be a field and $\mathfrak{m} \subset R = k[X_1, \dots, X_m]$ be a maximal ideal. Then $L = R/\mathfrak{m}$ is a finite extension field of k .

Proof: By Noether normalization lemma (1.1.6), there exist $a_1, \dots, a_n \in L$ algebraically independent such that L is integral over $k[a_1, \dots, a_n]$. By lemma (1.1.7), we conclude

that $k[a_1, \dots, a_n]$ is a field. Suppose that $n > 0$, so a_1 has an inverse. This means that $a_1 g(a_1, \dots, a_n) = 1$ for some $g \in k[X_1, \dots, X_n]$, which contradicts the fact that $a_1, \dots, a_n$ are algebraically independent. Then we conclude that L is an integral (and then algebraic) over k . However, since L is a finitely generated extension ($L = k(\overline{X}_1, \dots, \overline{X}_m)$) over k we conclude that L is in fact a finite field extension over k . $\square$

Theorem 1.1.10. Suppose that k is an algebraically closed field. Let $I \subset R = k[X_1, \dots, X_m]$ be an ideal, then $V(I) = \emptyset \iff I = k[X_1, \dots, X_m]$.

Proof: If $I \neq R$, then there exist a maximal ideal $\mathfrak{m} \supset I$. So $V(I) \supset V(\mathfrak{m})$ and it's sufficient to prove that $V(\mathfrak{m}) \neq \emptyset$. By theorem (1.1.9), we have that $L = R/\mathfrak{m}$ is a finite k -extension, so we must have $L = k$ because k is algebraically closed. So there exist $a_i \in k, i = 1, \dots, m$ such that $X_i - a_i \in \mathfrak{m}$ and $(X_1 - a_1, \dots, X_m - a_m) \subset \mathfrak{m}$. However $\varphi : R \rightarrow k : F \mapsto F(a_1, \dots, a_m)$ has kernel $\text{Ker}(\varphi) = (X_1 - a_1, \dots, X_m - a_m)$ and since k is a field, $R/\text{Ker}(\varphi)$ must be a field and $\text{Ker}(\varphi)$ is a maximal ideal. Therefore we conclude that $\mathfrak{m} = (X_1 - a_1, \dots, X_m - a_m)$. Finally $V(X_1 - a_1, \dots, X_m - a_m) = \{a_1, \dots, a_m\} \neq \emptyset$. $\square$

Remark 1.1.11. The hypothesis $k = \overline{k}$ in the previous theorem is crucial. For example if $k = \mathbb{R}$, we have $V(x^2 + 1) = \emptyset$.

Corollary 1.1.12. From the proof of the previous theorem, we see that there exist a bijection

$$\begin{aligned} k^n &\longleftrightarrow \left\{ \begin{array}{c} \text{maximal ideals} \\ \mathfrak{m} \subset R \end{array} \right\} \\ (a_1, \dots, a_n) &\longmapsto (X_1 - a_1, \dots, X_n - a_n) \end{aligned} \quad (1.7)$$

Theorem 1.1.13. (Hilbert Nullstellensatz). Let $A \subset k[X_1, \dots, X_n]$ be an ideal, then

$$IV(A) = \sqrt{A}. \quad (1.8)$$

Proof: The inclusion $\sqrt{A} \subset IV(A)$ is easy. Now, assume $g \in IV(A)$, where $A = (f_1, \dots, f_m)$. This means that we have the hypothesis:

$$f_i(a_1, \dots, a_n) = 0, \forall i = 1, \dots, m \Rightarrow g(a_1, \dots, a_n) = 0 \quad (*) \quad (1.9)$$

Now, define the ideal

$$I = Ak[X_1, \dots, X_{n+1}] + (1 - g(X_1, \dots, X_n)X_{n+1}). \quad (1.10)$$

If I is a proper ideal, then by corollary (1.1.12), we have $I \subset (X_1 - a_1, \dots, X_{n+1} - a_{n+1})$, which is the kernel of the evaluation map $k[X_1, \dots, X_{n+1}] \rightarrow k : f \mapsto f(a_1, \dots, a_{n+1})$. This means that

$$\begin{cases} f_i(a_1, \dots, a_n) = 0, \forall i = 1, \dots, m \\ g(a_1, \dots, a_n)a_{n+1} = 1 \end{cases} \quad (1.11)$$

which contradicts our initial hypothesis (*). Therefore, I is not a proper ideal, so $1 \in I$ and there exists $h_1, \dots, h_{m+1} \in k[X_1, \dots, X_{n+1}]$ such that

$$1 = \sum_{j=1}^m h_j(X_1, \dots, X_{n+1}) f_j(X_1, \dots, X_n) + h_{m+1}(X_1, \dots, X_{n+1})(1 - g(X_1, \dots, X_n)X_{n+1}). \quad (1.12)$$

Now, by projecting this equation onto $k[X_1, \dots, X_{n+1}]/(1 - gX_{n+1})$, we get

$$g^\ell(X_1, \dots, X_n) = \sum_{j=1}^m h_j(X_1, \dots, X_{n+1}) g^\ell \times f_j(X_1, \dots, X_n) \quad (1.13)$$

modulo $(1 - gX_{n+1})$, where $H_j = h_j(X_1, \dots, X_{n+1}) g^\ell \in k[X_1, \dots, X_n]$, $\forall j$ if $\ell \gg 0$. Since $\{\bar{X}_1, \dots, \bar{X}_n\}$ is algebraically free in $k[X_1, \dots, X_{n+1}]/(1 - gX_{n+1})$, we conclude that, in fact, we have equality

$$g^\ell(X_1, \dots, X_n) = \sum_{j=1}^m H_j(X_1, \dots, X_n) \times f_j(X_1, \dots, X_n) \quad (1.14)$$

in $k[X_1, \dots, X_n]$. □

Remark 1.1.14. We could have used the homomorphism

$$k[X_1, \dots, X_{n+1}] \rightarrow k(X_1, \dots, X_n) : X_{n+1} \mapsto g, \quad X_i \mapsto X_i, \quad i \neq n+1 \quad (1.15)$$

at the end of the previous proof.

1.2 Lecture 3

Definition 1.2.1. We define an affine algebraic set $V \subset k^n$ to be a set of the form $V = V(S)$ for some $S \subset k[\underline{X}]$.

Notation 1.2.2. Let V be an affine algebraic set. We write

$$D_V(g) = \{p \in V : g(p) \neq 0\} = V \setminus V(g). \quad (1.16)$$

These open sets are called “principal open subsets”.

Lemma 1.2.3. Any open subset $U \subset V$ of an affine algebraic set is quasicompact.

Proof: Consider a covering $U = \bigcup_{\lambda \in \Lambda} U_\lambda$. Observe that $V \setminus U = V(g_1, \dots, g_m)$, so we have the equality $U = \bigcup_{i=1}^m D_V(g_i)$. In particular, we see that principal open subsets form a basis for the Zariski topology on V . Therefore, it's sufficient to prove that a principal

open subset is quasicompact. So we assume $U = D_V(g)$. Furthermore, we have $U_\lambda = \bigcup_{i=1}^{n_\lambda} D_V(g_{\lambda,i})$ and

$$U = \bigcup_{\lambda \in \Lambda} \bigcup_{i=1}^{n_\lambda} D_V(g_{\lambda,i}). \quad (1.17)$$

So, without loss of generalization, we can assume $U = \bigcup_{\lambda \in \Lambda} D_V(g_\lambda)$ from beginning. Now

$$\begin{aligned} D_V(g) &= \bigcup_{\lambda \in \Lambda} D_V(g_\lambda) \iff \\ V \cap V(g) &= V \cap V(\langle g_\lambda : \lambda \in \Lambda \rangle) \iff \\ V \cap V(g) &= V \cap V(\langle g_\lambda : \lambda \in I \rangle) \quad (I \subset \Lambda \text{ finite}). \end{aligned} \quad (1.18)$$

In the last equation we used Hilbert basis theorem: The sequence

$$\langle g_{\lambda_1} \rangle \subset \langle g_{\lambda_1}, g_{\lambda_2} \rangle \subset \cdots \quad (1.19)$$

must terminate at some point, because $k[\underline{X}]$ is Noetherian. Then we conclude that $D_V(g) = \bigcup_{\lambda \in I} D_V(g_\lambda)$. $\square$

Lemma 1.2.4. Suppose that k is an infinity field, then open sets $U \subset k^n$ are dense in the Zariski topology.

Proof: Since basic open sets form a basis for the Zariski topology on k^n , it's sufficient to prove that $D(f)$ is dense in k^n . Let $\overline{D(f)} = V(g_1, \dots, g_m)$, $m \geq 1$ and $x \in k^n$, then we have for all $i = 1, \dots, m$:

$$f(x) \neq 0 \Rightarrow g_i(x) = 0 \Rightarrow fg_i(x) = 0, \forall x \in k^n. \quad (1.20)$$

Therefore, since k is infinity, we have $fg_i = 0 \in k[\underline{X}]$, $\forall i$ and $g_i = 0$, $\forall i$. Then we see that $\overline{D(f)} = k^n$. $\square$

Definition 1.2.5. (Regular functions). Let $V \subset k^n$ be an affine algebraic set and $U \subset V$ be an open set. A map $\varphi : U \rightarrow k$ is said to be *regular* if for any $p \in U$, there exist $p \in U' \subset U$, polynomials $f, g \in k[\underline{X}]$ such that $g(q) \neq 0$, $\forall q \in U'$ and $\varphi|_{U'} = f/g|_{U'}$.

Notation 1.2.6. Let V be an affine algebraic set and let $U \subset V$ be an open set, we denote $\mathcal{O}_V(U) = \{U \xrightarrow{\text{regular}} k\}$.

Lemma 1.2.7. Let $V \subset k^n$ be an affine algebraic set and $g \in k[X_1, \dots, X_n]$. Then the set of regular functions $\varphi : D_V(g) \rightarrow k$ is of the form f/g^m for some $f \in k[X_1, \dots, X_n]$ and $m > 0$. So

$$\mathcal{O}_V(D_V(g)) \cong \tilde{\psi}(k[X_1, \dots, X_n]_g) \quad (1.21)$$

where $\psi : k[X_1, \dots, X_n] \rightarrow \mathcal{O}_V(D_V(g))$ is the map that takes a polynomial and restricts it to $D_V(g)$ and $\tilde{\psi}$ is the corresponding map induced by the universal property of localization.

Proof: First we have $D_V(g) = \bigcup_{i=1}^{\ell} D_V(g_i)$ and since φ is regular, there exists $f_i, h_i \in k[\underline{X}]$, $i = 1, \dots, \ell$ such that

$$q \in D_V(g_i) \Rightarrow h_i(q) \neq 0 \text{ (so } q \in D_V(h_i)), \varphi|_{D_V(g_i)} = \frac{f_i}{h_i} \Big|_{D_V(g_i)}. \quad (1.22)$$

Now, we have two fundamental relations:

$$\begin{cases} D_V(g_i) \subset D_V(h_i), \forall i = 1, \dots, \ell \\ D_V(g) = \bigcup_{i=1}^{\ell} D_V(g_i) \end{cases} \quad (1.23)$$

Using $V = V(F_1, \dots, F_m)$ and Hilbert Nullstellensatz (1.1.13), we get

$$\begin{cases} \sqrt{(F_1, \dots, F_m, g_i)} \subset \sqrt{(F_1, \dots, F_m, h_i)}, \forall i = 1, \dots, \ell & (*) \\ \sqrt{(F_1, \dots, F_m, g)} = \sqrt{(F_1, \dots, F_m, g_1, \dots, g_{\ell})} & (**) \end{cases} \quad (1.24)$$

From (*), we deduce that $g_i^{n_i} = \sum_j A_j F_j + B_i h_i$ and restricting to V we get $g_i^{n_i} = B_i h_i$. Now, since $\varphi|_{D_V(g_i)} = \frac{B_i f_i}{g_i^{n_i}}$ and $D_V(g_i) = D_V(g_i^{n_i})$, we can **change notation** $B_i f_i \rightarrow f_i$ and $g_i^{n_i} \rightarrow g_i$.

Now we do a trick: since $g_j f_i - f_j g_i = 0$ on $D_V(g_i) \cap D_V(g_j) = D_V(g_i g_j)$, we get

$$g_i g_j (g_j f_i - f_j g_i) = 0 \text{ on } V. \quad (1.25)$$

Furthermore, we still have $\varphi|_{D_V(g_i)} = g_i f_i / g_i^2$. So we **change notation** again: $g_i^2 \rightarrow g_i$ and $g_i f_i \rightarrow f_i$, so we have

$$g_i f_j - g_j f_i = 0, \forall i, j = 1, \dots, \ell \quad (**). \quad (1.26)$$

Now, by equation (**), there exist $m > 0$ such that

$$g^m = \sum_j A_j F_j + \sum_k B_k g_k \quad (1.27)$$

and we get on V :

$$g^m = \sum_k B_k g_k. \quad (1.28)$$

Now, by using equation $(***)$, we have

$$f_i g^m = \sum_k B_k f_i g_k = \sum_k B_k f_k g_i = (\sum_k B_k f_k) g_i. \quad (1.29)$$

So if $q \in D_V(g_i)$, we have $\varphi(q) = \frac{f_i}{g_i}(q) = \frac{f}{g^m}(q)$, where we set $f = \sum_k B_k f_k$. $\square$

Corollary 1.2.8. Since $V = D_V(1)$, we have

$$\mathcal{O}_V(V) = \psi(k[X_1, \dots, X_n]) \cong k[\underline{X}]/I(V) = k[V]. \quad (1.30)$$

where $\psi : k[\underline{X}] \rightarrow \mathcal{O}_V(V)$ is as in the previous theorem.

Definition 1.2.9. A morphism between two affine algebraic sets $V_1 \subset k^n$ and $V_2 \subset k^m$ is a function $\varphi : V_1 \rightarrow V_2$ such that for any open set $U \subset V_2$, we have

$$\varphi_U^\# : \mathcal{F}(U, k) \xrightarrow{-\circ\varphi} \mathcal{F}(\varphi^{-1}(U), k), \quad \varphi_U^\#(\mathcal{O}_{V_1}(U)) \subset \mathcal{O}_{V_2}(\varphi^{-1}(U)). \quad (1.31)$$

Using more fancy words, the previous definition says that a continuous map $\varphi : V_1 \rightarrow V_2$ is a morphism if the composition map $\varphi_U^\# : \mathcal{F}(U, k) \xrightarrow{-\circ\varphi} \mathcal{F}(\varphi^{-1}(U), k)$ induces a sheaf morphism

$$\varphi^\#(U) : \mathcal{O}_{V_2} \rightarrow f_*\mathcal{O}_{V_1}. \quad (1.32)$$

We use the notation $f_*\mathcal{O}_{V_1}$ for direct image sheaf, it's defined by $f_*\mathcal{O}_{V_1}(U) = \mathcal{O}_{V_1}(f^{-1}(U))$.

1.3 Lecture 4

Definition 1.3.1. A function space is a topological space X with a sheaf $\mathcal{O}_X$ such that for any open set $U \subset X$, we have

- (1) $\mathcal{O}_X(U) \xrightarrow{\text{subring}} \{U \xrightarrow{\text{continuous}} k\}$
- (2) Let $f \in \mathcal{O}_X(U)$, then f has an inverse in $\mathcal{O}_X(U)$ if and only if $f(q) \neq 0, \forall q \in U$.

Lemma 1.3.2. An affine algebraic set $V \subset k^n$ equipped if the sheaf of regular functions is a function space.

Proof: Let $U \subset X$ be an open set and $f \in \mathcal{O}_X(U)$. Then, by definition, there exist an open cover $U = \bigcup_\lambda U_\lambda$ such that $f|_{U_\lambda} = f_\lambda/g_\lambda$, where $g_\lambda(q) \neq 0 \forall q \in U_\lambda$. To prove that f is continuous, it's sufficient to prove that $f^{-1}(a) \subset V$ is closed for all $a \in k$. So we prove that $(f|_{U_\lambda})^{-1}(a)$ is closed in V for all λ , which can be verified by the equality

$$V \cap V(f_\lambda - ag_\lambda) = (f|_{U_\lambda})^{-1}(a). \quad (1.33)$$

The second item is clear. $\square$

Definition 1.3.3. Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be two function spaces. We define a morphism $X \rightarrow Y$ to be a continuous map $f : X \rightarrow Y$ which induces a sheaf map $f^\# : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$ in the sense that if $f^\#(U)(g) = g \circ f$ for $g \in \mathcal{O}_Y(U)$, then $f^\#(U)(\mathcal{O}_Y(U)) \subset \mathcal{O}_X(f^{-1}(U))$.

Remark 1.3.4. Being a morphism of function spaces is a local property. In fact, let $f : X \rightarrow Y$ be a morphism of function spaces and suppose that $Y = \bigcup_\lambda V_\lambda$ and for each λ :

$$f|_{f^{-1}(V_\lambda)} : f^{-1}(V_\lambda) \rightarrow V_\lambda \quad (1.34)$$

is a morphism. Then f is a morphism.

Definition 1.3.5. An affine algebraic variety is a function space $(X, \mathcal{O}_X)$ isomorphic to an affine algebraic set with Zariski topology and sheaf of regular functions.

Definition 1.3.6. Let $(X, \mathcal{O}_X)$ be a function space, and $U \subset X$ be an open set with induced topology and $\mathcal{O}_U = \mathcal{O}_X|_U$. Then $(U, \mathcal{O}_U)$ is a function space called induced open subspace.

Definition 1.3.7. An abstract algebraic variety is a function space $(X, \mathcal{O}_X)$ such that there exist an open covering $X = \bigcup_\lambda U_\lambda$ with the property that for each λ , the induced open subspace $(U_\lambda, \mathcal{O}_{U_\lambda})$ is an affine algebraic variety.

Remark 1.3.8. In many texts we have the additional hypothesis of irreducibility in the definition of algebraic varieties.

Definition 1.3.9. An affine algebraic variety is said to be quasi-affine if it's isomorphic, as a function space, to an induced open subspace of an affine algebraic variety.

Lemma 1.3.10. Let $(X, \mathcal{O}_X)$ be a function space and $Y \subset k^n$ be a quasi affine algebraic set

Lemma 1.3.11. Let $(X, \mathcal{O}_X)$ be a function space and $Y \subset k^n$ be a quasi-affine algebraic set. Then $f : X \rightarrow Y$ is a morphism if and only if $f = (g_1, \dots, g_n)$ with $g_i \in \mathcal{O}_X(X)$. Here, we think Y as an induced open subspace of an affine algebraic set.

Proof: ($\Rightarrow$) The coordinate functions $x_i|_Y : Y \rightarrow k$ are regular functions, by definition. Now $g_i = x_i \circ f = f^\#(Y)(x_i) \in \mathcal{O}_X(f^{-1}(Y)) = \mathcal{O}_X(X)$, because $x_i \in \mathcal{O}_Y(Y) = \{Y \xrightarrow{\text{regular}} k\}$.

($\Leftarrow$) A closed set of Y is of the form $V(F) \cap Y$ (would not be of the form $V(F_1, \dots, F_m)$ in general ?). However $f^{-1}(V(F) \cap Y) = \{p \in X : F(g_1, \dots, g_n)(p) = 0\}$ which is closed in X provided we show that $F(g_1, \dots, g_n)$ is continuous. However, since $\mathcal{O}_X(X)$ is a ring, we have $F(g_1, \dots, g_n) \in \mathcal{O}_X(X)$. Furthermore, if $U \subset Y$ is an open set and $\varphi : U \rightarrow k$ is regular, we need to show that $f^\#(U)(\varphi) \in \mathcal{O}_X(f^{-1}(U))$. Let $U' \subset U$ be such that $\varphi|_{U'} = F/G$, $G(q) \neq 0 \forall q \in U'$. Then we have

$$f^\#(U')(\varphi) = \varphi \circ f|_{f^{-1}(U')} = \frac{F(g_1, \dots, g_n)}{G(g_1, \dots, g_n)} \Big|_{f^{-1}(U')}. \quad (1.35)$$

Furthermore

$$G \circ f(q) = G(g_1, \dots, g_n)(q) \neq 0, \forall q \in f^{-1}(U'). \quad (1.36)$$

Since $G(g_1, \dots, g_n)|_{f^{-1}(U')} \in \mathcal{O}_X(f^{-1}(U'))$, the previous equation shows that the inverse $1/G(g_1, \dots, g_n)$ also is an element of $\mathcal{O}_X(f^{-1}(U'))$. Now, let $U = \bigcup_\lambda U_\lambda$, $\varphi|_{U_\lambda} = F_\lambda/G_\lambda$, then we see that

$$f^\#(U_\lambda)(\varphi|_{U_\lambda}) \subset \mathcal{O}_X(f^{-1}(U_\lambda)). \quad (1.37)$$

The glueing axiom shows that $f^\#(U)(\varphi) \in \mathcal{O}_X(f^{-1}(U))$. $\square$

Lemma 1.3.12. Let $V \subset k^n$ be an affine algebraic set and $D_V(F) \subset V$ an affine algebraic set. Then the induced open subspace $(D_V(F), \mathcal{O}_{D_V(F)})$ is a quasi-affine algebraic variety, by definition. However, we have more, it's also an affine algebraic variety.

Proof: Let $V = V(G_1, \dots, G_m)$ and define the map

$$\varphi : D_V(F) \rightarrow V(x_0 F - 1, G_1, \dots, G_m) \subset k^{n+1} : (a_1, \dots, a_n) \mapsto \left(\frac{1}{F(a_1, \dots, a_n)}, a_1, \dots, a_n \right) \quad (1.38)$$

By the lemma (1.3.11), we see that φ is a morphism. Furthermore, it has a set theoretical inverse given by the projection

$$\psi : V(x_0F - 1, G_1, \dots, G_m) \rightarrow D_V(F) : (a_0, \dots, a_n) \mapsto (a_1, \dots, a_n). \quad (1.39)$$

By the same lemma, this map is also a morphism. Since $f\psi = \mathbb{1}_{D_V(x_0F-1, \dots)}$ and $\psi f = \mathbb{1}_{D_V(F)}$, we see that φ is an isomorphism (the induced sheaf morphism would be an isomorphism by construction). $\square$

Remark 1.3.13. Let $(X, \mathcal{O}_X)$ be an affine algebraic variety and $\varphi : (V, \mathcal{O}_V) \rightarrow (X, \mathcal{O}_X)$ an isomorphism, where $V \subset k^n$ is an algebraic set. Then this isomorphism, induces another isomorphism

$$(D_V(F), \mathcal{O}_{D_V(F)}) \rightarrow (X_f, \mathcal{O}_{X_f}) \quad (1.40)$$

where $F = f \circ \varphi$ and $X_f = \varphi(D_V(F)) = \{p \in X : f(p) \neq 0\}$.

Corollary 1.3.14. A quasi-affine algebraic variety is an abstract algebraic variety.

Proof: A quasi-affine algebraic variety has a topological basis consisting of principal open subsets, which are affine algebraic varieties.

1.4 Lecture 5 and 6

Theorem 1.4.1. Suppose that we have the data:

- $(X_\lambda)_{\lambda \in \Lambda}$ function spaces over k
- $\forall \lambda, \mu \in \Lambda$ let $U_{\lambda\mu} \subset X_\lambda$
- $\forall \lambda, \mu \in \Lambda$ an isomorphism

$$\varphi_{\lambda\mu} : U_{\lambda\mu} \rightarrow U_{\mu\lambda} \quad (1.41)$$

Furthermore, assume that

- $U_{\lambda\lambda} = X_\lambda, \varphi_{\lambda\lambda} = 1_{X_\lambda}$
- $\forall \lambda, \mu, \nu \in \Lambda$, we have

$$\begin{aligned} \varphi_{\lambda\mu}(U_{\lambda\mu} \cap U_{\lambda\nu}) &\subset U_{\mu\lambda} \cap U_{\mu\nu} \\ \varphi_{\mu\nu}\varphi_{\lambda\mu}|_{U_{\lambda\mu} \cap U_{\lambda\nu}} &= \varphi_{\lambda\nu}|_{U_{\lambda\mu} \cap U_{\lambda\nu}} \end{aligned} \quad (1.42)$$

Then, $X = \coprod_{\lambda \in \Lambda} X_\lambda / \sim$, where we put $(\lambda, x) \sim (\mu, x')$ if and only if $\varphi_{\lambda\mu}(x) = x'$. One can verify that this defines an equivalence relation. Furthermore, X has a topology defined as the small topology such that all the maps $i_\lambda : X_\lambda \rightarrow X : x \mapsto [(\lambda, x)]$ are continuous. We define a sheaf

$$\mathcal{O}_X(U) = \{f : U \rightarrow k : fi_\lambda \in \mathcal{O}_{X_\lambda}(i_\lambda^{-1}(U)), \forall \lambda \in \Lambda\}. \quad (1.43)$$

With this sheaf, $(X, \mathcal{O}_X)$ is a function space satisfying the following universal property: for any function space Y and morphisms $j_\lambda : X_\lambda \rightarrow Y$ such that:

$$j_\mu \varphi_{\lambda\mu} = j_\lambda|_{U_{\lambda\mu}}, \quad (1.44)$$

there exist a unique morphism $f : X \rightarrow Y$ such that $fi_\lambda = j_\lambda$ (morphisms here, means morphisms of function spaces!). Furthermore, since $i_\mu^{-1}i_\lambda(X_\lambda) = U_{\mu\lambda} \subset X_\mu, \forall \mu$, then $i_\lambda(U_\lambda)$ is open and i_λ are isomorphisms onto its images. So, if each X_λ is an affine algebraic variety, then X is an abstract variety.

Proof: Tedious.

Definition 1.4.2. An abstract variety X is separable if for any pair of affine open sets $U, V \subset X$, we have

- (1) $U \cap V$ is affine
- (2) The induced map

$$\mathcal{O}(U) \otimes \mathcal{O}(V) \rightarrow \mathcal{O}(U \cap V) : f \otimes g \mapsto f|_{U \cap V} \cdot g|_{U \cap V} \quad (1.45)$$

is surjective.

Example 1.4.3. Let $X_0 = X_1 = k^1$ and $U_{01} = U_{10} = k^1 - \{0\}$. Define a map $\varphi_{01} : U_{01} \rightarrow U_{10} : x \mapsto 1/x$. We claim that $X = \mathbb{P}^1$ in the variety obtained by glueing X_0 and X_1 through φ_{01} . In fact, for $i = 0, 1$, let $g_i : X_i \rightarrow Y$ be two morphisms such that

$$g_1 \varphi_{01} = g_0|_{U_{01}}. \quad (1.46)$$

That is, $g_1(1/x) = g_0(x)$. Then the following map is well defined:

$$\begin{aligned} g : X &\rightarrow Y \\ [1 : y] &\mapsto g_0(y) \\ [x : 1] &\mapsto g_1(x) \end{aligned} \quad (1.47)$$

Moreover, by construction we have $g|_{U_i} = g_i \varphi_i$, where $\varphi_i : U_i \rightarrow k^1$ (here, $U_i = \{[x_0, x_1] : x_i \neq 0\}$) are the standard charts of the projective plane X . So we see that g is locally a morphism, i.e., it's a morphism.

Is X separable? Consider the following diagram:

$$\begin{array}{ccccc} \mathcal{O}_X(U_0) \otimes \mathcal{O}_X(U_1) & \xrightarrow{\quad} & \mathcal{O}_X(U_0 \cap U_1) & & \\ \uparrow & & \downarrow & & \\ \mathcal{O}_{k^1}(k^1) \otimes \mathcal{O}_{k^1}(k^1) & & \mathcal{O}_{k^1}(k^1 - \{0\}) & & \\ & \nearrow \varphi_0^\#(s) \otimes \varphi_1^\#(t) & & \xrightarrow{\quad} & \varphi_0^\#(s)|_{U_0 \cap U_1} \varphi_1^\#(t)|_{U_0 \cap U_1} \\ & \uparrow & & & \downarrow \\ & s \otimes t & \dashrightarrow & & (\varphi_0^{-1})^\#(\varphi_0^\#(s)|_{U_0 \cap U_1} \varphi_1^\#(t)|_{U_0 \cap U_1}) \end{array}$$

So the composition $\mathcal{O}_{k^1}(k^1) \otimes \mathcal{O}_{k^1}(k^1) \rightarrow \mathcal{O}_{k^1}(k^1 - \{0\})$ yields

$$s \otimes t \mapsto s \cdot t \circ \varphi_1 \circ \varphi_0^{-1}|_{k^1 - \{0\}} \quad (1.48)$$

At the level of polynomial rings (see lemma (1.2.7)), this map is

$$k[x] \otimes k[y] \rightarrow k[u]_u \cong k[u, u^{-1}] : f \otimes g \mapsto f(u)g(u^{-1}) \quad (1.49)$$

which is clearly surjective. **TODO:** finish the argument.

1.5 Lecture 6,7,8

Remark 1.5.1. Take the product topology on $k^1 \times k^1$, where k^1 is equipped with Zariski topology. Let $U = k^1 - S$ and $V = k^1 - T$ (S, T finite) be two open sets. Then $U \times V = k^1 \times k^1 - (S \times k^1 \cup k^1 \times T)$ and in general, for a family of open sets U_λ, V_λ , we have the equality.

$$\begin{aligned} (\bigcup_{\lambda} U_{\lambda} \times V_{\lambda})^c &= \bigcap_{\lambda} S_{\lambda} \times k^1 \cup k^1 \times T_{\lambda} \\ &= (\bigcap_{\lambda} S_{\lambda} \times k^1) \cup (k^1 \times \bigcap_{\lambda} T_{\lambda}) \end{aligned} \quad (1.50)$$

where S_{λ}, T_{λ} are all finite. So any closed set $F \subset k^1 \times k^1$ is of the form

$$F = S \times k^1 \cup k^1 \times T, \#S, \#T < \infty \quad (1.51)$$

in the product topology. In particular, if $\#k = \infty$, then $V(X, Y) = \{0\}$ is not closed in the product topology.

TODO: remains to study lect 8, Nullstellensatz for projective spaces and “universal compactness” theorem.

1.5.1 Segre embedding

Remark 1.5.2. Here, to ease notation, we will write $x = [x_0, \dots, x_m]$ sometimes.

The Segre embedding is the map

$$s : \mathbb{P}^m \times \mathbb{P}^n \rightarrow \mathbb{P}^{(m+1)(n+1)-1}$$

$$([x_0, \dots, x_m], [y_0, \dots, y_n]) \mapsto \begin{bmatrix} x_0 y_0 & x_0 y_1 & \dots & x_0 y_n \\ x_1 y_0 & x_1 y_1 & \dots & x_1 y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_m y_0 & x_m y_1 & \dots & x_m y_n \end{bmatrix} \quad (1.52)$$

where we parametrize $\mathbb{P}^{(m+1)(n+1)-1}$ with projective coordinates z_{ij} , $0 \leq i \leq m, 0 \leq j \leq n$. We claim that $s(\mathbb{P}^m \times \mathbb{P}^n)$ is a closed subset. In fact, if we let $z_{ij} = x_i y_j$, then we have $z_{ij} z_{pq} = z_{iq} z_{pj}$ and

$$s(\mathbb{P}^m \times \mathbb{P}^n) \subset V = V_p(z_{ij} z_{pq} - z_{iq} z_{pj}, \forall p, q, i, j) \quad (*). \quad (1.53)$$

Conversely, if we take $z \in V$, then $z_{ij} \neq 0$ for some i, j . So we assume $z_{ij} = 1$. Then we see that $z_{pq} = z_{pj}z_{iq}$, so we can take $x_p = z_{pj}$ and $y_q = z_{iq}$ to conclude that $s(x, y) = z$. Then we conclude the equality of $(*)$. Furthermore, we claim that this map is injective. Let $s(x, y) = s(x', y')$, then $x_p y_q = \lambda x'_p y'_q$. Fix q such that $y_q \neq 0$, then $x_p = (\frac{\lambda y'_q}{y_q}) x'_p$, $\forall p$ and $x = x'$. In a similar way we see that $y = y'$.

As a corollary, we see that s mapst $\mathbb{P}^m \times \mathbb{P}^n$ bijectively to a closed subset $V \subset \mathbb{P}^{(m+1)(n+1)-1}$. We put a structure of abstract variety on $\mathbb{P}^m \times \mathbb{P}^n$ in such a way that s is an isomorphism onto V . So we see for example, that

$$\mathcal{O}_{\mathbb{P}^m \times \mathbb{P}^n}(U) = (- \circ s)(\mathcal{O}_V(s(U))). \quad (1.54)$$

In a concrete way, a section $s \in \mathcal{O}_{\mathbb{P}^m \times \mathbb{P}^n}$ is a quotient of bihomogeneous polynomials of bi degree (d, d) each one, for some $d \geq 0$. In fact, if $F(z) \in k[\underline{z}]$ is homogeneous, then

$$F(z) = \sum a_\alpha \prod_{i,j} z_{ij}^{\alpha_{ij}}, \quad (i, j) \in \{0, \dots, m\} \times \{0, \dots, n\}. \quad (1.55)$$

Therefore

$$\begin{aligned} F \circ s(x, y) &= \sum a_\alpha \prod_{i,j} x_i^{\alpha_{ij}} y_j^{\alpha_{ij}} \\ &= \sum a_\alpha (\prod_i x_i^{\sum_j \alpha_{ij}}) (\prod_j y_j^{\sum_i \alpha_{ij}}) \end{aligned} \quad (1.56)$$

where $\sum_i \sum_j \alpha_{ij} = \sum_j \sum_i \alpha_{ij} = |\alpha| = d$, the degree of F . It's easy to see that a quotient of bihomogeneous polynomials of bidegree (d, e) each one, is also a section of $\mathcal{O}_{\mathbb{P}^m \times \mathbb{P}^n}$. In fact, let G/H be a quotient of two bihomogeneous polynomials of bidegree (d, e) each one. Then, in $D(x_i) \times D(y_j)$ we have

$$\frac{G}{H} = \frac{G x_i^e y_j^d}{H x_i^e y_j^d}. \quad (1.57)$$

So we can assume both G, H to be of the same bidegree $(d + e, d + e)$. Moreover, let F be a bihomogeneous polynomial of bidegree (d, d) , then

$$\begin{aligned} F(x, y) &= \sum a_{\alpha\beta} x^\alpha y^\beta \\ &= \sum a_{\alpha\beta} x_{i_1} y_{j_1} \cdots x_{i_d} y_{j_d} \quad (1 \leq i_k \leq m, 1 \leq j_k \leq n) \\ &= \sum a_{\alpha\beta} z_{i_1 j_1} \cdots z_{i_d j_d} \end{aligned} \quad (1.58)$$

is homogeneous of degree d .

Lemma 1.5.3. The variety $\mathbb{P}^m \times \mathbb{P}^n$ constructed via the Segre embedding is a product.

Proof: Let $f : Z \rightarrow \mathbb{P}^m$ and $g : Z \rightarrow \mathbb{P}^n$ be morphisms. We draw the following diagram

$$\begin{array}{ccccc}
& & D(x_i) \times D(y_j) & \xrightarrow{s} & D(z_{ij}) \cap V \\
& \nearrow (f,g)|_U & \downarrow (\varphi_i, \psi_j) & \nearrow s \circ (\varphi_i^{-1}, \psi_j^{-1}) & \\
U = f^{-1}(D(x_i)) \cap g^{-1}(D(y_j)) & & k^m \times k^n & & \\
& \searrow (\varphi_i f|_U, \psi_j g|_U) & & &
\end{array}$$

We see that $\varphi_i f|_U, \psi_j g|_U$ are morphisms, so $(\varphi_i f|_U, \psi_j g|_U)$ is a morphism because it's a mapping into a product of affine varieties and we can apply the affine criteria. So, if we show that $s \circ (\varphi_i^{-1}, \psi_j^{-1})$ is a morphism, we are done. However, we see that

$$\xi_{ij} \circ s \circ (\varphi_i^{-1}, \psi_j^{-1}) : k^m \times k^n \rightarrow k^{(m+1)(n+1)} : (x, y) \mapsto (\dots, x_i, \dots, y_j, \dots, x_i y_j, \dots) \quad (1.59)$$

which is clearly a morphism. $\square$

Remark 1.5.4. The above proof also shows that $(\varphi_i, \psi_j) : D(x_i) \times D(y_j) \rightarrow k^m \times k^n$ is a morphism. Furthermore, let $X \subset \mathbb{P}^m, Y \subset \mathbb{P}^n$ be two closed subsets. Then, is $X \cap D(x_i) \times Y \cap D(y_j) \subset D(x_i) \times D(y_j)$ closed? We have $\varphi_i(X \cap D(x_i)) \subset k^m, \psi_j(Y \cap D(y_j)) \subset k^n$ are closed. Therefore, the product

$$\varphi_i(X \cap D(x_i)) \times \psi_j(Y \cap D(y_j)) = (\varphi_i, \psi_j)(X \cap D(x_i) \times Y \cap D(y_j)) \quad (1.60)$$

is closed in $k^m \times k^n$. Here we use the standard observation that $V(S) \times V(T) = V(S \cup T)$, where $S \subset k[\underline{X}], T \subset k[\underline{y}]$ with both rings sitting inside $k[\underline{x}, \underline{y}]$. Since (φ_i, ψ_j) is a homeomorphism, this proves that the answer is positive.

Corollary 1.5.5. Let $X \subset \mathbb{P}^m, Y \subset \mathbb{P}^n$ be two closed subsets. Then $X \times Y \subset \mathbb{P}^m \times \mathbb{P}^n$ is closed.

Remark 1.5.6. The product of the affine spaces k^m and k^n if $k^m \times k^n \cong k^{m+n}$. So, if $f : Z \rightarrow k^m, g : Z \rightarrow k^n$ are morphisms, then we draw the diagram.

$$\begin{array}{ccccc}
& & k^m \times k^n & & \\
& \nearrow (f,g) & \downarrow p_1 & \searrow p_2 & \\
Z & & k^m & & k^n \\
& \searrow f & \downarrow g & \nearrow & \\
& & & &
\end{array}$$

However, f being a morphism means that $x_i \circ f$ and $y_j \circ g$ are morphisms for all i, j ($1 \leq i \leq m, 1 \leq j \leq n$), where x_i and y_j are the standard coordinate functions. However, the set of coordinate functions of the product $k^m \times k^n$ is $(x_1, \dots, x_m, y_1, \dots, y_n)$ because of the identification $k^m \times k^n \cong k^{m+n}$. So, again we see that

$$(x_1 \circ f, \dots, x_m \circ f, y_1 \circ g, \dots, y_n \circ g) = (f, g) \quad (1.61)$$

is a morphism. Here we used lemma (1.3.11) several times.

1.5.2 Grassmanians and Flag varieties

Let $\text{Gr}_d(V)$ be the set of d -dimensional subspaces of V . We have the following embedding

$$\psi : \text{Gr}_d(V) \hookrightarrow \mathbb{P}(\bigwedge^d V) : W = \text{Span}(w_1, \dots, w_d) \mapsto [w_1 \wedge \dots \wedge w_d]. \quad (1.62)$$

The map is well defined, in fact, if $\{w_i\}_{i=1}^d$ and $\{w'_i\}_{i=1}^d$ are two basis of W and $w'_i = \sum_j b_{ij} w_j$, then $w'_1 \wedge \dots \wedge w'_d = \det(b_{ij})_{i,j=1}^d w_1 \wedge \dots \wedge w_d$. Now, let $v_1, \dots, v_n$ be a basis of V and $u_1, \dots, u_d \in V$ be arbitrary vectors. Write $u_i = \sum_j a_{ij} v_j$, then

$$\begin{aligned} u_1 \wedge \dots \wedge u_d &= \left(\sum_{j_1=1}^n a_{1j_1} v_{j_1} \right) \wedge \dots \wedge \left(\sum_{j_d=1}^n a_{dj_d} v_{j_d} \right) \\ &= \sum_{j_1=1}^n \dots \sum_{j_d=1}^n a_{1j_1} \dots a_{dj_d} v_{j_1} \wedge \dots \wedge v_{j_d} \\ &= \sum_{j_1 < \dots < j_d} \det(a_{ij_k})_{i,k=1}^d v_{j_1} \wedge \dots \wedge v_{j_d} \end{aligned} \quad (1.63)$$

So we see that $u_1 \wedge \dots \wedge u_d = 0 \iff \text{Rank}(a_{ij}) < d \iff \{u_1, \dots, u_d\}$ is linearly dependent. So, if

$$w_1 \wedge \dots \wedge w_d = \lambda w'_1 \wedge \dots \wedge w'_d \neq 0, \quad (1.64)$$

then $u \wedge w_1 \wedge \dots \wedge w_d = 0 \iff u \wedge w'_1 \wedge \dots \wedge w'_d = 0$, i.e., $u \in \text{Span}\{w_i\}_{i=1}^d \iff u \in \text{Span}\{w'_i\}_{i=1}^d$. Now, let $\mathcal{J} = \{(j_1, \dots, j_d) : 1 \leq j_1 < \dots < j_d \leq n\}$ and if $J = (j_1, \dots, j_d) \in \mathcal{J}$, write $v_J = v_{j_1} \wedge \dots \wedge v_{j_d}$. Then we have the basis $\{v_J\}_{J \in \mathcal{J}}$ of $\bigwedge^d V$ and we write X_J for the corresponding coordinate functions.

Let $W \in \text{Gr}_d(V)$, we see that $\psi(W) \in D(X_{(1, \dots, d)})$ if and only if $\det(a_{ij})_{i,j=1}^d \neq 0$. In this case, by row reduction, we can assume that $a_{ij} = \delta_{ij}$, $(1 \leq i, j \leq d)$, so W has a basis consisting of vectors of the form $w_i = v_i + \sum_{j>d} a_{ij} v_j$, $i = 1, \dots, d$. Now define J_{pq} as the tuple $(j_1, \dots, j_d)$ such that $j_k = k$ ($k < p$), $j_k = k + 1$, ($p \leq k < d$) and $j_d = q \in \{d + 1, \dots, n\}$. With this notation, we see that

$$a_{pq} = \pm \det(a_{ij_k})_{j,k=1}^d = \pm X_{J_{pq}}(w_1 \wedge \dots \wedge w_d). \quad (1.65)$$

So, for any $J = (j_1, \dots, j_d) \in \mathcal{J}$, the expression $\det(a_{ij_k})_{i,k=1}^d$ is a polynomial $f_J(X) \in k[X_{J_{pq}} : 1 \leq p \leq d, d < q \leq n] \subset k[X_J : J \in \mathcal{J} \setminus \{(1, \dots, d)\}]$ and we see that

$$\begin{aligned} \varphi_{(1, \dots, d)}(\text{Im}(\psi) \cap D(X_{(1, \dots, d)})) \\ = V(X_J - f_J(X) : \forall J = (j_1, \dots, j_d) \in \mathcal{J} \setminus \{(1, \dots, d)\}, J_{pq} : p, q) \end{aligned} \quad (1.66)$$

which is closed, so $\text{Im}(\psi) \cap D(X_{(1, \dots, d)})$ is closed. With the same argument applied to the other principal open subsets $D(X_J)$, we see that $\text{Im}(\psi)$ is closed. It's important to note that the row echelon form of $A = (a_{ij})$ gives an affine coordinate expression for $\psi(W) \in U_{(1, \dots, d)}$:

$$\varphi_{(1, \dots, d)}\psi(W) = (f_J(a))_{J \in \mathcal{J} \setminus \{(1, \dots, d)\}} \quad (1.67)$$

where $f_J(a) = \det(a_{ij_k})_{i,k=1}^d$, $J = (j_1, \dots, j_d)$.

Now we define the Flag variety $\mathcal{F}(V)$, it's defined as the n -tuple of subspaces

$$(F_1, \dots, F_n) \in \text{Gr}_1(V) \times \dots \times \text{Gr}_n(V) \quad (1.68)$$

such that $F_i \subset F_{i+1}$, $1 \leq i < n$. It's a closed subset, so it's a projective variety in the above product. We prove this in two steps. The first one is to prove that

$$\{(F, F') : F \subset F'\} \subset \text{Gr}_d(V) \times \text{Gr}_{d+1}(V) \quad (1.69)$$

is a closed subset. As before, we simplify the problem by showing that the intersection

$$\{(F, F') : F \subset F'\} \cap U_{(1, \dots, d)} \times U_{(1, \dots, d, d+1)} \quad (*) \quad (1.70)$$

is closed. So let F and F' be represented by matrices $(a_{ij}) \in \text{Mat}_{d,n}(k)$ and $(b_{pq}) \in \text{Mat}_{d+1,n}(k)$ in row echelon form, i.e., $a_{ij} = \delta_{ij}$, $j \leq d$ and $b_{pq} = \delta_{pq}$, $q \leq d+1$. In this condition $F \subset F'$ if and only if the matrix

$$M = \begin{pmatrix} b_{11} & \dots & b_{1n} \\ \vdots & \dots & \vdots \\ b_{d+1,1} & \dots & b_{d+1,n} \\ a_{11} & \dots & a_{1n} \\ \vdots & \dots & \vdots \\ a_{d1} & \dots & a_{dn} \end{pmatrix} \quad (1.71)$$

has rank $d+1$. In particular, this means that any $d+2$ minor of M must vanish, which is a polynomial condition on the coefficients a_{ij}, b_{pq} . By equation (1.65), we see that this defines a polynomial condition in the affine coordinate system of $\varphi_{(1, \dots, d)} \times \varphi_{(1, \dots, d+1)}$, hence intersection $(*)$ is closed.

Now, for the second step, we use corollary (1.5.5) to conclude that

$$S_d = \text{Gr}_1(V) \times \dots \times \text{Gr}_{d-1}(V) \times \{(F, F') \in \text{Gr}_d(V) \times \text{Gr}_{d+1}(V) : F \subset F'\} \times \dots \times \text{Gr}_{d+2}(V) \times \dots \times \text{Gr}_n(V) \quad (1.72)$$

is closed. Therefore

$$\mathcal{F}(V) = \bigcap_{d=1}^{n-1} S_d \quad (1.73)$$

is closed, so it's a projective variety.

Lemma 1.5.7. We have $\dim(\text{Gr}_d(V)) = d(n-d)$, $n = \dim(V)$.

Proof: From equation (1.66), we see that at $U_{(1, \dots, d)} \subset \mathbb{P}(\wedge^d V)$, the Grassmanian $\text{Gr}_d(V)$ is defined by $V(I)$, where

$$I = (X_J - f_J(X) : \forall J = (j_1, \dots, j_d) \in \mathcal{J} \setminus \{(1, \dots, d), J_{pq} : p, q\}) \quad (1.74)$$

Put any LEX order on $k[X_J : J \in \mathcal{J} \setminus \{(1, \dots, d)\}]$, with the property $X_J > X_{J_{pq}}$ whenever $J \notin \{J_{pq} : (p, q) \in \{1, \dots, d\} \times \{d+1, \dots, n\}\}$. Then, we see that $\sqrt{I} = I$ and

$$\frac{k[X_J : J \neq (1, \dots, d)]}{I} \cong k[X_{J_{pq}} : (p, q) \in \{1, \dots, d\} \times \{d+1, \dots, n\}] \quad (1.75)$$

which has transcendence degree $d(n-d)$. $\square$

Remark 1.5.8. There is a simple way to represent an element $F \in \mathcal{F}(V)$. Fix a basis $\{e_1, \dots, e_n\}$ of V and observe that we can find a basis of V such that $F_i = \langle v_1, \dots, v_i \rangle$. Now, assume that $F \in U_{(1)} \times \dots \times U_{(1, \dots, d-1)} \times U_{(1, \dots, d)}$. Then if we write $v_i = \sum_j a_{ij} e_j$, we see that $A = (a_{ij})$ has nonsingular principal minors M_{ii} , $i = 1 \dots, n$. We call A a representative matrix of F .

Since $a_{11} \neq 0$, we can assume $v_1 = e_1 + \sum_{j>1} a_{1j} e_j$. Now, observe that if we only do elementary operations of the form $R_j \rightarrow R_j + kR_i$, ($i < j$) and $R_j \rightarrow kR_j$, ($k \neq 0$) (where $R_i = (a_{i1}, \dots, a_{in})$ is the i th row of A) on A , we still have A as a representative of the flag F . Using the hypothesis that the principal minors are nonsingular, we see that we can assume $a_{ii} = 1, \forall i$, while $a_{ij} = 0$, ($i > j$). That is $v_i = e_i + \sum_{j>i} a_{ij} e_j$.

1.6 Lecture 9

Chapter 2

Algebraic Groups

This is some notes that i did based on the first chapter of the book [1].

2.1 Basic definitions

Some important observations:

- If $\mathbb{F}$ is a finite field, the map which takes a polynomial in $\mathbb{F}[X_1, \dots, X_n]$ and send to a function $\mathbb{F}^n \rightarrow \mathbb{F}$ is not an isomorphism. For example, if $\mathbb{F}$ has q elements, then $\prod_{q \in \mathbb{F}} (X - q) \in \mathbb{F}[X] \mapsto 0$. However, if $\mathbb{F}$ is infinity and $a_n X^n + a_{n-1} X^{n-1} + \dots + a_0 = 0$, then we can select distinc elements $\alpha_0, \dots, \alpha_n$ and we conclude that

$$\begin{pmatrix} a_0 & \dots & a_n \end{pmatrix} \begin{pmatrix} 1 & 1 & \dots & 1 \\ \alpha_0 & \alpha_1 & \dots & \alpha_n \\ \vdots & \vdots & \dots & \vdots \\ \alpha_0^n & \alpha_1^n & \dots & \alpha_n^n \end{pmatrix} = 0. \quad (2.1)$$

However $\det(\alpha_i^j)_{i,j=0}^n = \prod_{i < j} (\alpha_i - \alpha_j) \neq 0$ and we conclude that $a_0, \dots, a_n = 0$. Now, if $f \in \mathbb{F}[X_1, \dots, X_n]$, then we can write (assumin X_1 appears in f):

$$f = g_d(X_2, \dots, X_n)X_1^d + \dots + g_1(X_2, \dots, X_n)X_1 + g_0(X_2, \dots, X_n). \quad (2.2)$$

So the equality $f = 0$, the previous step for the case $\mathbb{F}[X]$ and an induction on the number of variables yields $g_d, \dots, g_0 = 0$. Then we see that if $|\mathbb{F}| = \infty$, we can identify the ring $\mathbb{F}[X_1, \dots, X_n]$ with polynomial functions $\mathbb{F}^n \rightarrow \mathbb{F}$.

As a consequence of this story, we see that if $|\mathbb{F}| = \infty$, then

$$I(\mathbb{F}^n) = (0) \subset \mathbb{F}[X_1, \dots, X_n]. \quad (2.3)$$

- (Hilbert basis theorem). The Hilbert basis theorem states that if A is noetherian, then $A[X]$ is noetherian. Since $\mathbb{F}[X]$ is a PID (and in consequence noetherian), we conclude by an induction on the number of variables that $\mathbb{F}[X_1, \dots, X_n]$ is noetherian.

Remark 2.1.1. A ring is said to be noetherian if it satisfy the ascending chain condition, or equivalently, if all of its ideals are finitely generated.

- (Algebraic sets). Let $S \subset \mathbb{F}[X_1, \dots, X_n]$, then $V(S) = \{a \in \mathbb{F}^n : f(a) = 0\}$. It's immediate that $V(S) = V(\langle S \rangle)$. The relation $V(\bigcup_{i \in I} S_i) = \bigcap_{i \in I} V(S_i)$ allow us to define the Zariski topology on $\mathbb{F}^n$. Moreover, for any subset $V \subset \mathbb{F}^n$, we let $I(V) = \{f \in \mathbb{F}[X_1, \dots, X_n] : f(a) = 0, \forall a \in V\}$. Then we have

$$\begin{cases} VI(W) = \overline{W} \\ IV(\mathfrak{a}) = \sqrt{\mathfrak{a}} \end{cases} \quad (2.4)$$

We also have the important relation

$$V(\mathfrak{a} \cap \mathfrak{b}) = V(\mathfrak{a}\mathfrak{b}) = V(\mathfrak{a}) \cup V(\mathfrak{b}). \quad (2.5)$$

In fact

$$V(\mathfrak{a}) \cup V(\mathfrak{b}) \underset{\mathfrak{a} \cap \mathfrak{b} \subset \mathfrak{a}, \mathfrak{b}}{\subset} V(\mathfrak{a} \cap \mathfrak{b}) \underset{\mathfrak{a}\mathfrak{b} \subset \mathfrak{a} \cap \mathfrak{b}}{\subset} V(\mathfrak{a}\mathfrak{b}) = V(\{fg : f \in \mathfrak{a}, g \in \mathfrak{b}\}) \subset V(\mathfrak{a}) \cup V(\mathfrak{b}). \quad (2.6)$$

- (Coordinate rings). Let $\mathfrak{a} = I(V)$ where V is an algebraic set. We define the coordinate ring $A[V] = \mathbb{F}[X_1, \dots, X_n]/\mathfrak{a}$.
- (LEX order). We define an ordering on $\mathbb{Z}_{\geq 0}^n$ by $\alpha > \beta$ if for the smallest i such that $\alpha_i \neq \beta_i$ we have $\alpha_i > \beta_i$. So, with a fixed LEX on $\mathbb{F}[X_1, \dots, X_n]$, we can define the leading term of a polynomial: $\text{LT}(f) = a_\alpha X^\alpha$, where α is the smallest tuple, relative to LEX such that $a_\alpha \neq 0$. In the case $\text{LT}(f) = a_\alpha X^\alpha$, we also write $\text{LM}(f) = X^\alpha$.
- (Groebner Basis). Let $\mathfrak{a}$ be an ideal in $\mathbb{F}[X_1, \dots, X_n]$. A groebner basis of $\mathfrak{a}$ is a finite subset $G \subset \mathfrak{a}$ such that $(\text{LT}(g) : g \in G) = \text{LT}(\mathfrak{a})$.

Example 2.1.2. Let $R = \mathbb{F}[X_1, \dots, X_n, Y_1, \dots, Y_n]$ with LEX: $X_1 > \dots > X_n > Y_1 > \dots > Y_n$ and define an ideal $\mathfrak{a} = (X_1 - g_1, \dots, X_n - g_n)$ for $g_i \in \mathbb{F}[Y_1, \dots, Y_n]$. Since $\text{LT}(X_i - g_i) = X_i$, it's clear that $(X_1, \dots, X_n) \subset \text{LT}(\mathfrak{a})$. Now, if $f \in R$ and $X_j \nmid \text{LT}(f)$, $\forall j$, then $f \in \mathbb{F}[Y_1, \dots, Y_n]$ because its smallest term has power $\alpha \in \{0\} \times \mathbb{Z}^n$. So, we need to show that $I \cap \mathbb{F}[Y_1, \dots, Y_n] = (0)$. This can be seen by proving that the kernel of $\varphi : R \rightarrow \mathbb{F}[Y] : X_i \mapsto g_i, Y_i \mapsto Y_i$ is I . The only nontrivial inclusion is $\text{Ker}(\varphi) \subset I$. To see this use a division algorithm to write

$$f = h_1(X_1 - g_1) + \dots + h_n(X_n - g_n) + r \quad (2.7)$$

where $r \neq 0$ implies that no term of r is divisible by $\text{LT}(X_i - g_i) = X_i$, $\forall i$. By applying φ on both sides of this equation we have the desired inclusion. From this we see that $G = (X_1 - g_1, \dots, X_n - g_n)$ is a groebner basis for I .

- (Hilbert Polynomial).

Definition 2.1.3. Let $I \subset \mathbb{F}[X_1, \dots, X_n]$ be an ideal. We define the affine Hilbert function:

$${}^a\text{HF}_I : \mathbb{Z}_{>0} \rightarrow \mathbb{Z}_{>0} : s \mapsto \dim_{\mathbb{F}} \left(\frac{\mathbb{F}[X_1, \dots, X_n]_{\leq s}}{I_{\leq s}} \right). \quad (2.8)$$

Example 2.1.4. For example, the number ${}^a\text{HF}_{(0)}(s)$ is the number of n -tuples of nonnegative integers whose sum is $\leq s$. To compute this, write the generating function

$$\prod_{i=1}^n \frac{1}{1 - X_i} = \sum_{\alpha \in \mathbb{Z}_{\geq 0}^n} X^\alpha. \quad (2.9)$$

Now put $X_i = t$, $\forall i$. We see that

$$(1 - t)^{-n} = \sum_{s \geq 0} \binom{-n}{s} (-1)^s t^s = \sum_{s \geq 0} \binom{n+s}{s} t^s. \quad (2.10)$$

So ${}^a\text{HF}_{(0)}(s) = \binom{n+s}{s} = \frac{1}{n!} s^n + \dots$.

Example 2.1.5. Let $f \in \mathbb{F}[X_1, \dots, X_n]$ be a degree d polynomial. Then

$${}^a\text{HF}_{(f)}(s) = \binom{s+n}{s} - \binom{s-d+n}{s-d} = \frac{d}{(n-1)!} s^{n-1} + \dots. \quad (2.11)$$

An important property of the affine Hilbert function is the following:

Theorem 2.1.6. Let $I \subset \mathbb{F}[X_1, \dots, X_n]$ be an ideal, then

$${}^a\text{HF}_I(s) = {}^a\text{HF}_{\text{LT}(I)}(s), \quad s \geq 0. \quad (2.12)$$

Another main result is the existence of the Hilbert polynomial.

Theorem 2.1.7. There exist a polynomial ${}^a\text{HP}_I(t) \in \mathbb{Q}[t]$ such that

$${}^a\text{HP}_I(s) = {}^a\text{HF}_I(s), \quad \forall s \geq 0. \quad (2.13)$$

Proof (sketch):

- Find a Groebner basis G of I
- Let $M \subset \mathbb{Z}_{\geq 0}^n$ be such that $\{X^\beta\}_{\beta \in M} = \{\text{LM}(g) : g \in G\}$
- Write

$$C(I) = \bigcap_{\beta \in M} C(\beta), \quad C(\beta) = \{\alpha \mid X^\beta \nmid X^\alpha\} \quad (2.14)$$

as a union of $C(J, \tau) = \{\alpha \mid \alpha_j = \tau(j) \forall j \in J\}$. To do this use

$$C(\beta) = \bigcup_{i=1}^n \bigcup_{t_i=0}^{\beta_i-1} C(\{i\}, \tau : i \mapsto t_i). \quad (2.15)$$

- Use the formula $|C(J, \tau)_{\leq s}| = \binom{d+s-|\tau|}{d}$, $d = n - |J|$, $|\tau| = \sum_{j \in J} \tau(j)$ and the inclusion-exclusion principle to compute

$${}^a\text{HP}_I(s) = |C(I)_{\leq s}| = \left| \bigcup_{(J, \tau)} C(J, \tau)_{\leq s} \right|. \quad (2.16)$$

Example 2.1.8. Let $I = (X_1 - X_3^2, X_2 - X_3^3)$ and fix the LEX order with $X_1 > X_2 > X_3$. Then we calculate

- $G = \{X_1 - X_3^2, X_2 - X_3^3\}$
- $M = \{(1, 0, 0), (0, 1, 0)\}$
- We have

$$\begin{aligned} C(I) &= C(\{1\}, \tau : 1 \mapsto 0) \cap C(\{2\}, \tau : 2 \mapsto 0) \\ &= C(\{1, 2\}, \tau : 1 \mapsto 0, 2 \mapsto 0). \end{aligned} \quad (2.17)$$

So we conclude that $|C(I)|_{\leq s} = \binom{1+s-0}{1} = s + 1$.

- (Algebraic dimension). Here we define the dimension of an algebraic set.

Definition 2.1.9. Let $V \subset \mathbb{F}^n$ be an algebraic set. We define its dimension as $\dim(V) = \deg {}^a\text{HP}_{I(V)}(t)$. It has the property that it's the biggest integer $d \geq 1$ such that there exist indexes $1 \leq i_1 < \dots < i_d \leq n$ such that $I(V) \cap \mathbb{F}[X_{i_1}, \dots, X_{i_d}] = (0)$.

Example 2.1.10. Let $H_f = V(f)$, f irreducible, then

$$\begin{aligned} \dim(H_f) &= \deg {}^a\text{HP}_{I(H_f)}(t) \\ &= \deg {}^a\text{HP}_{(f)}(t) \\ &= \deg\left(\frac{\deg(f)}{(n-1)!}t^{n-1} + \dots\right) \\ &= n - 1. \end{aligned} \quad (2.18)$$

The dimension has the expected property: If $W \subset V$ is closed, then $\dim(W) \leq \dim(V)$. In fact

$$\begin{aligned} I(V) \subset I(W) &\Rightarrow \\ I(V)_{\leq s} \subset I(W)_{\leq s} &\Rightarrow \\ {}^a\text{HF}_{I(W)}(s) \leq {}^a\text{HF}_{I(V)}(s), \forall s \geq 0 &\Rightarrow \\ \deg {}^a\text{HF}_{I(W)}(t) \leq \deg {}^a\text{HF}_{I(V)}(t). \end{aligned} \quad (2.19)$$

- (Transcendence degree). Let A be a field which is also a finitely generated $\mathbb{F}$ -algebra. We define the transcendence degree of A over $\mathbb{F}$ (and write $\partial_{\mathbb{F}}(A)$) to be largest size of an algebraically independent subset of A over $\mathbb{F}$.

Example 2.1.11. We have $\partial_{\mathbb{F}}(\mathbb{F}[X_1, \dots, X_n]) = n$. Indeed, let $\{f_1, \dots, f_r\}$ be polynomials and suppose that $r > n$. Write

$$\begin{aligned} W_s &= \{f^\beta : \beta \in \mathbb{Z}_{\geq 0}^r, |\beta| \leq s\}, \quad d = \max\{\deg(f_i) : i = 1, \dots, r\} \\ V_s &= \{X^\alpha : |\alpha| \leq s\} \end{aligned} \quad (2.20)$$

Then $W_s \subset V_{ds}$ and

$$|W_s| = \binom{r+s}{s} \stackrel{s \gg 0}{\sim} \binom{n+sd}{sd} = \dim(V_{sd}) \geq \dim(W_s). \quad (2.21)$$

So, for $s \gg 0$ there exist $\{a_\beta\}_\beta \subset \mathbb{F}$ such that $\sum_\beta a_\beta f^\beta = 0$ and $\{f_1, \dots, f_r\}$ is algebraically dependent.

Theorem 2.1.12. Let V be an algebraic set and write $I = I(V)$, then

$$\dim(V) = \partial_{\mathbb{F}}(A[V]) = \partial_{\mathbb{F}}\left(\frac{\mathbb{F}[X_1, \dots, X_n]}{I}\right). \quad (2.22)$$

Proof: Let $d = \dim(V)$, so there exist indexes $1 \leq i_1 < \dots < i_d \leq n$ such that $I \cap \mathbb{F}[X_{i_1}, \dots, X_{i_d}] = (0)$. This is equivalent to say that $[X_{i_1}], \dots, [X_{i_d}] \in A[V]$ are algebraically independent over $\mathbb{F}$. Therefore $d \leq \partial_{\mathbb{F}}(A[V])$. Now, suppose that $\varphi_i = [f_i] \in A[V], i = 1, \dots, m$ are algebraically independent over $\mathbb{F}$ and put $N = \max \deg(f_i)$, then the map

$$\mathbb{F}[Y_1, \dots, Y_m]_{\leq s} \mapsto \frac{\mathbb{F}[X_1, \dots, X_n]_{\leq Ns}}{I_{\leq Ns}} : g \mapsto g(\varphi_1, \dots, \varphi_m) \quad (2.23)$$

is injective. Then, by taking dimensions we get

$$\binom{m+s}{s} \leq {}^a\text{HF}_I(s), \quad \forall s \geq 0. \quad (2.24)$$

We know that both expressions are polynomials in s , so by taking degrees we conclude that $m \leq \dim(V) = d$ and $\partial_{\mathbb{F}}(A[V]) \leq d$. $\square$

With small changes in the above proof, we can verify that in the case of $A[V]$ being an integral domain, we have also $\partial_{\mathbb{F}}(K) = \dim(V)$, $K = \text{Frac}(A[V])$.

Example 2.1.13. Let $C = V(X_1 - X_3^2, X_1 - X_3^3)$, then $I(C) = (X_1 - X_3^2, X_2 - X_3^3)$ and

$$\frac{\mathbb{F}[X_1, X_2, X_3]}{I(C)} \cong \mathbb{F}[X_3] \quad (2.25)$$

so $\partial_{\mathbb{F}}(A[C]) = 1$.

Example 2.1.14. The purpose of this example is to show that if $V \subset \mathbb{F}^n, |V| = \infty$ is a linear space, then $\dim(V) = \dim_{\mathbb{F}}(V)$.

- Let $\text{Ann}(V) = \text{Span}(f_1, \dots, f_m)$. Since V has finite dimension, we have $v \in V \iff f(v) = 0 \forall f \in \text{Ann}(V)$ and $V = V(\{f_1, \dots, f_m\})$.
- Let $f_i = \sum a_{ij} X_j$, where we identify X_j with the j th coordinate functions in $\mathbb{F}^n$. Now, if $A = (a_{ij})$ and apply reduction to put A in the form:

$$\begin{pmatrix} a_{1j_1} & * & 0 & * & \cdots & 0 & \cdots \\ & & a_{2j_2} & * & \cdots & 0 & \cdots \\ & & & & \ddots & \cdots & \vdots \\ & & & & & 0 & \cdots & a_{mj_m} & \cdots \end{pmatrix} \quad (2.26)$$

with $a_{ij_i} = 1$, $j_1 < j_2 < \dots$. Now put LEX with: $X_{j_1} > \dots > X_{j_m} > X_j$ $j \notin \{j_1, \dots, j_m\}$ and write $f_i = X_{j_i} + h_i$ with $h_i \in \mathbb{F}[X_j : j \notin \{j_1, \dots, j_m\}]$. Then we see that $A[V] \cong \mathbb{F}[X_j : j \notin \{j_1, \dots, j_m\}]$ and $(f_1, \dots, f_m)$ is prime. Therefore $I(V) = (f_1, \dots, f_m)$. Finally

$$\dim(V) = \dim_{\mathbb{F}}(A[V]) = \dim_{\mathbb{F}} \mathbb{F}[X_j : j \notin \{j_1, \dots, j_m\}] = \dim_{\mathbb{F}} V. \quad (2.27)$$

2.2 Tangent Spaces

- (Regular maps). Let $V \subset \mathbb{F}^m, W \subset \mathbb{F}^n$ be algebraic sets. Then a map $f : V \rightarrow W$ is said to be regular if there exist polynomials such that $f = (f_1, \dots, f_m)$. Such a map is continuous in the Zariski topology.

Remark 2.2.1. We can identify $A[V]$ with the set of regular functions $V \rightarrow \mathbb{F}$. In this context, we have a contravariant functor:

$$\varphi : V \rightarrow W \rightsquigarrow \varphi^* : A[W] \rightarrow A[V] : [g] \mapsto [g] \circ \varphi \quad (2.28)$$

Example 2.2.2. Let $\varphi : \mathbb{F} \rightarrow C : x \mapsto (x^3, x^2, x)$ and write $C = (X_1 - X_3^2, X_2 - X_3^3)$. Remember the isomorphism $A[C] \rightarrow \mathbb{F}[Y] : [X_1] \mapsto Y^2, [X_2] \mapsto Y^3, [X_3] \mapsto Y$. So we have

$$\varphi^* : A[C] \rightarrow A[\mathbb{F}] : [X_3] \mapsto [X_3] \circ \varphi. \quad (2.29)$$

The first map is $[X_3] : C \rightarrow \mathbb{F} : (x^3, x^2, x) \mapsto x$ and the last map is $\mathbb{F} \rightarrow C : x \mapsto x$. At the algebraic level, the map φ^* is just the isomorphism $\mathbb{F}[Y] \cong \mathbb{F}[X]$.

- (Tangent spaces). Let V be an algebraic set and $V_p = \mathbb{F}$ be the $A[V]$ -module with action $[f]\alpha \equiv f(p)\alpha$. Let $I(V) = (f_1, \dots, f_m)$. We define

$$\begin{aligned} T_p(V) &= \{v \in \mathbb{F}^n : d(f)_p(v) = 0 \forall f \in I(V)\} \\ &= \{v \in \mathbb{F}^n : d(f_i)_p(v) = 0 \forall i\} \\ &= V(\{d(f_i)_p : i = 1, \dots, m\}) \end{aligned} \quad (2.30)$$

Remark 2.2.3. Caution ! The polynomials of f_i here are not the generators of V , but of $I(V)$. For example, if $V = V(f)$, we can conclude that $T_p(V) = V(df_p)$ only if f is irreducible, i.e., if $I(V) = (f)$.

Here $df_p \equiv \sum_{i=1}^n D_i(f)(p)X_i \in \mathbb{F}[X_1, \dots, X_n]$. We have an isomorphism

$$T_p(V) \rightarrow \text{Der}_{\mathbb{F}}(A[V], V_p) : v \mapsto D_v; \quad D_v([f]) = \sum_{i=1}^n [D_i(f)] \cdot v_i \in \mathbb{F}. \quad (2.31)$$

with inverse $D \mapsto (D([X_i]))_{i=1}^n$. Of course $D_v([f]) = df_p(v)$.

Maybe it will be better to change notation of V_p to ${}^p\mathbb{F}$ or something.

Example 2.2.4. Let $C = (f_1, f_2)$, $f_1 = X_1 - X_3^2$, $f_2 = X_2 - X_3^3$ and $p = (x^3, x^2, x) \in C$. We have

$$d(f_1)_p = 1 - 2xX_3, \quad d(f_2)_p = 1 - 3x^2X_3 \quad (2.32)$$

and $T_p(C) = \mathbb{F}\langle (3x^2, 2x, 1) \rangle$.

Example 2.2.5. The determinant is an irreducible polynomial. Indeed, let $\det \in \mathbb{F}[X_{ij} : 1 \leq i, j \leq n]$. Then, if we apply Laplace theorem on the first row, we get

$$\det = A_{11}X_{11} - A_{12}X_{12} + \dots + (-1)^n A_{1n}X_n \quad (2.33)$$

where A_{ij} denotes the determinant of the matrix obtained by excluding the i th row and j th column. By induction on the number of variables, we assume that A_{ij} are irreducible polynomials. Now, suppose that $\det = fg$, where $\deg(f), \deg(g) > 0$. Choose LEX with $X_{ij} > X_{k\ell}$ if $i < k$ or if $i = k$ and $j < \ell$. Then

$$\text{LT}(A_{11})X_{11} = \text{LT}(\det) = \text{LT}(f)\text{LT}(g). \quad (2.34)$$

Assume that $X_{11} | \text{LT}(f)$ (so $\text{LT}(g) < X_{11}$) and write $f = AX_{11} + B$, $\text{LT}(B) < X_{11}$. Then we must have $AgX_{11} = A_{11}X_{11}$ and $A_{11} = Ag$. Since $\deg(g) > 1$ and A_{11} is irreducible, we must have $g = A_{11}$ and $A = 1$. Now we get

$$-A_{12}X_{12} + \dots + (-1)^n A_{1n}X_n = BA_{11} \quad (2.35)$$

and $-\text{LT}(A_{12})X_{12} = \text{LT}(B)\text{LT}(A_{11})$. Since $X_{22} | \text{LT}(A_{11})$, we are led to $X_{22} | \text{LT}(A_{12})$ which is a contradiction because X_{22} doesn't appear in A_{12} by definition.

Now, since $\det$ is an irreducible homogeneous polynomial, the difference $\det - 1$ must be irreducible. Therefore $I(SL_n \mathbb{F}) = (\det - 1)$ and we can compute the tangent space

$T_I(SL_n) = V(d_I(\det))$. We have

$$\begin{aligned}
d_I \det &= \sum_{k,\ell} D_{k\ell}(\det)(I) X_{k\ell} \\
&= \sum_{k,\ell} \sum_{\sigma \in S_n} D_{k\ell} \prod_{i=1}^n X_{i\sigma(i)}(I) X_{k\ell} \\
&= \sum_{k,\ell} \sum_{\sigma \in S_n} \delta_{\ell,\sigma(k)} \prod_{i \neq k} \delta_{i,\sigma(i)} X_{k\ell} \\
&= \sum_{k,\ell} \delta_{k\ell} X_{k\ell} \quad (\delta_{\ell,\sigma(k)} \prod_{i \neq k} \delta_{i,\sigma(i)} \neq 0 \text{ only if } \sigma = 1) \\
&= \sum_k X_{kk}.
\end{aligned} \tag{2.36}$$

Therefore $T_I SL_n(\mathbb{F}) = \mathfrak{sl}_n \mathbb{F}$, the Lie algebra of traceless matrices.

- (Gauss lemma). We state the Gauss lemma.

Lemma 2.2.6. Let R be a UFD ring (think in $\mathbb{F}[X]$), then $R[X]$ is a UFD.

Proof: Let f and $K = \text{Frac}(R)$. Since $K[X]$ is a UFD, we have $f = FG$, $F, G \in K[X]$ with F irreducible and $\deg(G) < \deg(f)$. Since R is a UFD, there exist $r, s \in R$ such that $rF, sG \in R[X]$. Decompose $rs = t_1 \cdots t_k$ into irreducible elements. Then if $F' = rF, G' = sG$, we have

$$[F'][G'] = [rs][f] = 0 \in R/(t_i)[X] \tag{2.37}$$

because $f \in R[X]$. So, since $R/(t_i)[X]$ is a domain, we see that $t_i | F'$ or $t_i | G'$. Then we see that there exist a partition $\{1, \dots, k\} = I \cup J$ such that $F' / \prod_{i \in I} t_i, G' / \prod_{j \in J} t_j \in R[X]$. It's clear that F' still irreducible, then an induction on degree yields the result. $\square$

Corollary 2.2.7. Let $f, g \in R = \mathbb{F}[X_1, \dots, X_n]$ and assume f irreducible with $f \nmid g$, then there exist $F, G \in R$ such that

$$Gf + Fg = h \in \mathbb{F}[X_1, \dots, X_{n-1}] \setminus 0. \tag{2.38}$$

Proof: Let $K = \text{Frac}(\mathbb{F}[X_1, \dots, X_{n-1}])$, then $f, g \in K[X_n]$ and there exist $A, B \in K[X_n]$ such that $Af + Bg = 1$ (by Gauss lemma f still irreducible as an element of $K[X_n]$). Then there exist $h \in \mathbb{F}[X_1, \dots, X_{n-1}]$ such that $hA, hB \in \mathbb{F}[X_1, \dots, X_{n-1}]$. So by writing $G = hA, F = hB$ we are done. $\square$

Notation 2.2.8. Let $A = \mathbb{F}[X_1, \dots, X_n]/I$, where M is a A -module. We write

$$\mathcal{T}_{A,M} = \{(v_1, \dots, v_n) \in M^n : \sum_i [D_i(f)] \cdot v_i, \forall f \in I\}. \tag{2.39}$$

We have the isomorphism $\mathcal{T}_{A,M} \rightarrow \text{Der}_{\mathbb{F}}(A, M) : v \mapsto D_v$ with inverse

$$D \mapsto (D(\overline{X_i}))_{i=1}^n. \quad (2.40)$$

Now consider the following context, let $K = \text{Frac}(A)$ as A -module (A acts by multiplication). Then we have an isomorphism

$$\text{Der}_{\mathbb{F}}(K, K) \rightarrow \text{Der}_{\mathbb{F}}(A, K) : D \mapsto D|_A. \quad (2.41)$$

In particular we also have the isomorphism $\mathcal{T}_{A,K} \cong \text{Der}_{\mathbb{F}}(K, K)$. We observe that this is also a K -vector space isomorphism, it's a bijection compatible with the K -action. From this isomorphism, we see that $\mathcal{T}_{A,K}$ only depends on K , and the fact that $\text{Frac}(A) = K$.

Theorem 2.2.9. Let $A = \mathbb{F}[X_1, \dots, X_n]/I$ be a domain and $K = \text{Frac}(A)$. Then

$$\partial_{\mathbb{F}}(A) = \dim_K \text{Der}_{\mathbb{F}}(K, K). \quad (2.42)$$

Proof: Let $d = \partial_{\mathbb{F}}(A) = \partial_{\mathbb{F}}(K)$, then there exist $z_1, \dots, z_{d+1}$ such that

- $K = \mathbb{F}(z_1, \dots, z_{d+1})$
- $\{z_1, \dots, z_d\}$ is algebraically independent
- z_d is separable over $\mathbb{F}(z_1, \dots, z_d)$

Let $R = \mathbb{F}[z_1, \dots, z_{d+1}] \subset K$ and $f \neq 0$ irreducible such that $f(z_1, \dots, z_{d+1}) = 0$. Then by corollary (2.2.7), we see that $\alpha : K[Y_1, \dots, Y_{d+1}] \rightarrow K : g \mapsto g(z_1, \dots, z_{d+1})$ has kernel (f) . Therefore

$$\begin{aligned} \mathcal{T}_{R,K} &= \{(v_1, \dots, v_{d+1}) \in K^{d+1} : \sum_i [D_i(f)] \cdot v_i = 0\} \\ &= \{(v_1, \dots, v_{d+1}) \in K^{d+1} : \sum_i D_i(f)(z_1, \dots, z_{d+1}) \cdot v_i = 0\}. \end{aligned} \quad (2.43)$$

We see that $D_i(f)(z_1, \dots, z_{d+1}) = 0 \Rightarrow D_i(f) \in (f)$ and $D_i(f) = 0$. Since $f \neq 0$ and $|\mathbb{F}| = \infty$, we must have $D_i(f) \neq 0$ for some i , so $D_i(f)(z_1, \dots, z_{d+1}) \neq 0$ for some i . Then we conclude that

$$\dim_K \text{Der}_{\mathbb{F}}(K, K) = \dim_K \mathcal{T}_{R,K} = d = \partial_{\mathbb{F}}(K) = \partial_{\mathbb{F}}(A). \quad (2.44)$$

□

Corollary 2.2.10. If $A = \mathbb{F}[X_1, \dots, X_n]/(f_1, \dots, f_m)$, then

$$\partial_{\mathbb{F}}(A) = \dim_K \mathcal{T}_{A,K} = n - \text{Rank}_K([D_i(f_j)]). \quad (2.45)$$

Example 2.2.11. Let $C = V(f_1, f_2)$, $f_1 = X_1 - X_3^2$, $f_2 = X_2 - X_3^3$. We have

$$A[C] = \frac{\mathbb{F}[X_1, X_2, X_3]}{(f_1, f_2)} \rightarrow \mathbb{F}[X_3] : [g] \mapsto g(X_3^3, X_3^2, X_3). \quad (2.46)$$

So $A[C]$ is a domain and we can define $K = \text{Frac}(A[C])$. We compute

$$\text{Rank}_K([D_i(f_j)]) = \text{Rank}_K \begin{pmatrix} 1 & 0 & -2[X_3] \\ 0 & 1 & -3[X_3] \end{pmatrix} = 2. \quad (2.47)$$

Then we conclude $\dim(V) = 3 - 2 = 1$.

- (differential of a regular map). Here we define the notion of differential of a regular map.

Definition 2.2.12. Let $\varphi : V \rightarrow W$ be a regular map between algebraic varieties. Also, let $p \in V, q = \varphi(p) \in W$. We define:

$$d\varphi_p : \text{Der}_{\mathbb{F}}(A[V], V_p) \rightarrow \text{Der}_{\mathbb{F}}(A[W], W_q) : D_v \mapsto D_v \circ \varphi^* \quad (2.48)$$

At the level of tangent spaces, we have:

$$\begin{array}{ccc} \text{Der}_{\mathbb{F}}(A[V], V_p) & \xrightarrow{d\varphi_p} & \text{Der}_{\mathbb{F}}(A[W], W_q) \\ \uparrow & & \downarrow \\ T_p(V) & \xrightarrow{\quad \quad} & T_q(W) \\ & \uparrow \text{red} & \downarrow \text{red} \\ & v & (D_v \circ \varphi^*([Y_i]))_{i=1}^m \end{array}$$

(Note: The diagram shows a commutative square with red arrows. The top horizontal arrow is $d\varphi_p$. The left vertical arrow is an inclusion. The right vertical arrow is $D_v \mapsto D_v \circ \varphi^*$. The bottom horizontal arrow is $v \mapsto (D_v \circ \varphi^*([Y_i]))_{i=1}^m$. The middle vertical arrows are inclusions $T_p(V) \hookrightarrow \text{Der}_{\mathbb{F}}(A[V], V_p)$ and $T_q(W) \hookrightarrow \text{Der}_{\mathbb{F}}(A[W], W_q)$. The red arrows indicate the mapping from the tangent space level to the derivation level via the differential map.)

In more details, if we write $\varphi = (f_1, \dots, f_m)$, then $\varphi^*([Y_i]) = [f_i]$ and we get

$$D_v([f_i]) = \sum_{j=1}^m [D_j(f_i)] \cdot v_j = \sum_{j=1}^m D_j(f_i)(p) v_j. \quad (2.49)$$

Therefore

$$d\varphi_p(v) = (D_j(f_i)(p))_{i,j} \cdot v = J_p(\varphi) \cdot v. \quad (2.50)$$

Example 2.2.13. Let $G \subset \mathbb{F}^{n^2}$ be an algebraic group and $\mu : G \times G \rightarrow G$ be the product of matrices. We have $G \times G \subset \mathbb{F}^{n^2} \times \mathbb{F}^{n^2} \cong \mathbb{F}^{2n^2}$. Now, identify the first n^2

coordinate functions with X_{ij} and the second n^2 coordinate functions with Y_{ij} . So we have

$$\mu_{ij} = \sum_{\ell} X_{i\ell} Y_{\ell j}. \quad (2.51)$$

Then we have

$$\frac{\partial \mu_{ij}}{\partial X_{rs}}(I) = \frac{\partial \mu_{ij}}{\partial Y_{rs}}(I) = \delta_{ir} \delta_{sj}. \quad (2.52)$$

Therefore

$$d_{(I,I)}\mu(A, B) = \left(\sum_{r,s} \frac{\partial \mu_{ij}}{\partial X_{rs}}(I) A_{rs} + \frac{\partial \mu_{ij}}{\partial Y_{rs}}(I) B_{rs} \right)_{ij} = (A_{ij} + B_{ij}) = A + B. \quad (2.53)$$

TODO:

1. Prove that T_1G is a Lie algebra. Remember the star operator.
2. Compute the Lie algebra of SL_n using the fact that $\det -1$ is an irreducible polynomial.

Chapter 3

List of exercises

Exercise 1. Let $U = k^2 - \{0\}$, show that the set of regular functions $U \rightarrow k$ are polynomials, then conclude that U is not an affine variety.

Solution. Let $\varphi \in \mathcal{O}_U(U)$ be a regular function, then $\varphi|_{D_{k^2}(X)} = f/X^m$ and $\varphi|_{D_{k^2}(Y)} = g/Y^n$ and on the intersection $D_{k^2}(X) \cap D_{k^2}(Y) = D_{k^2}(XY)$, we have

$$Y^n f - X^m g = 0 \quad (*). \quad (3.1)$$

Since there exist an infinity number of points $p \in D_{k^2}(XY)$ (assuming k is algebraically closed), we also have the equality $(*)$ in $k[X, Y]$. Now, let $f = \sum a_{ij} X^i Y^j$ and $g = \sum b_{ij} X^i Y^j$. Then, equality $(*)$ means

$$a_{i,j-n} = a_{i-m,j}, \quad \forall i \geq m, j \geq n. \quad (3.2)$$

Therefore $f/X^m = g/Y^n \in k[X, Y]$ and φ is a polynomial, because $U = D_{k^2}(X) \cup D_{k^2}(Y)$. In conclusion, we proved that the inclusion $i : U \hookrightarrow k^2$ induces a ring isomorphism

$$i^\#(k^2) : k[X, Y] \rightarrow \mathcal{O}_U(U) : f \mapsto f|_U (**). \quad (3.3)$$

Now assume $U = V(S)$, $S \subset k[X, Y]$, then $f \in S \Rightarrow f|_U = 0$ and $f = 0$ because of $(**)$. So $S = \{0\}$ and $U = k^2$, which is a contradiction. $\square$

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